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List of Publications

JOURNAL PUBLICATIONS

(Accredited Journals. ATLAS/CERN collaboration papers listed below as indexed on Google Scholar.)

Non-ATLAS Journal Papers

1. Khanyisani Mhlangano Makhanya and Muaaz Bhamjee and Neil Martinson and Simon Connell, "Assessment and characterization of the impact of pulmonary pathology on flow-induced acoustics using computational fluid dynamics", *Biomedical Signal Processing and Control*, vol. 120, pp. 110018, 2026. DOI
2. Ralijaona, Mbolahasina and Igumbor, Emmanuel and Bhamjee, Muaaz and Otwombe, Kennedy and Nabeemeeah, Firdaus and Milovanovich, Minja and Martinson, Neil and Mafa, Pedro and Leeuw, Lerothodi and Connell, Simon, "Infectiousness model of expelled droplets exposed to ultraviolet germicidal irradiation coupled with evaporation", *Computers & Fluids*, vol. 275, pp. 106242, 2024. DOI
3. Mokoena, Fortune and Bhamjee, Muaaz and Connell, Simon H. and Van Vaerenbergh, Pierre and Iori, Gianluca and Kaprolat, Axel, "An FEA Investigation of the Vibration Response of the BEATS Detector Stage", *R&D Journal of the South African Institution of Mechanical Engineering*, vol. 39, pp. 44–52, 2023. DOI
4. Makhanya, K. M. and Connell, S. H. and Bhamjee, Muaaz and Martinson, N. A., "The Use of Computational Fluid Dynamics for Assessing Flow-Induced Acoustics to Diagnose Lung Conditions", *Mathematical and Computational Applications*, vol. 28, No. 3, pp. 64, 2023. DOI
5. Bhamjee, Muaaz and Connell, Simon H. and Nel, André Leon, "The Modification of the Dynamic Behaviour of the Cyclonic Flow in a Hydrocyclone under Surging Conditions", *Mathematical and Computational Applications*, vol. 27, No. 6, pp. 88, 2022. DOI
6. Potgieter, A. and Bhamjee, M. and Kruger, S., "Modelling of a Heated Gas-solid Fluidised Bed using Eulerian Based Models", *R&D Journal of the South African Institution of Mechanical Engineering*, vol. 37, pp. 45–57, 2021. DOI
7. Potgieter, M. S. W. and Bester, C. R. and Bhamjee, M., "Experimental and CFD Investigation of a Hybrid Solar Air Heater", *Solar Energy*, vol. 195, pp. 413–428, 2020. Impact Factor: 5.742. DOI
8. Coetzee, Rigardt Alfred Maarten and Jen, Tien-Chien and Bhamjee, Muaaz and Lu, Junling, "The mechanistic process analysis and temperature effect in a low pressure square type atomic layer deposition reactor", *International Journal of Mechanical Engineering and Robotics Research*, vol. 8, No. 5, pp. 792–796, 2019. DOI
9. Coetzee, Rigardt Alfred Maarten and Jen, Tien-Chien and Bhamjee, Muaaz and Lu, Junling, "The mechanistic effect over the substrate in a square type atomic layer deposition reactor", *International Journal of Modern Physics B*, vol. 33, No. 01n03, pp. 1940018, 2019. DOI
10. Bhamjee, M. and Nurick, A. and Madyira, D. M., "An Experimentally Validated Mathematical and CFD Model of a Supply Air Window: Forced and Natural Flow", *Energy and Buildings*, vol. 57, pp. 289–301, 2013. Impact Factor: 5.879. DOI

ATLAS/CERN Collaboration Papers

Note: Papers on which Bhamjee is a contributing author, as indexed on Google Scholar.

1. ATLAS Collaboration, "Observation of pair production of longitudinally polarized Z bosons in four-lepton final state with the ATLAS detector", *Physical Review Letters*, 2026. DOI

2. ATLAS Collaboration, "Search for boosted low-mass resonances decaying into hadrons produced in association with a photon in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector", *Journal of High Energy Physics*, vol. 2025, No. 1, pp. 99, 2025.
3. ATLAS Collaboration, "Measurements of W^+W^- production cross-sections in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector", *Journal of High Energy Physics*, vol. 2025, No. 8, pp. 1–66, 2025.
4. ATLAS Collaboration, "The performance of missing transverse momentum reconstruction and its significance with the ATLAS detector using 140 fb^{-1} of $\sqrt{s} = 13$ TeV pp collisions", *The European Physical Journal C*, vol. 85, No. 6, pp. 606, 2025.
5. ATLAS Collaboration, "Differential cross-section measurements of Higgs boson production in the $H \to \tau^+\tau^-$ decay channel in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector", *Journal of High Energy Physics*, vol. 2025, No. 3, pp. 1–66, 2025.
6. ATLAS Collaboration, "Search for diphoton resonances in the 66 to 110 GeV mass range using pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector", *Journal of High Energy Physics*, vol. 2025, No. 1, pp. 53, 2025. DOI
7. ATLAS Collaboration, "Observation of Υ Production in Pb+Pb Collisions at $\sqrt{s_{\mathrm{NN}}}$ with the ATLAS Detector", *Physical Review Letters*, vol. 134, No. 14, 2025.
8. ATLAS Collaboration, "Exploration at the high-energy frontier: ATLAS Run 2 searches investigating the exotic jungle beyond the Standard Model", *Physics Reports*, vol. 1116, pp. 301–385, 2025.
9. ATLAS Collaboration, "The quest to discover supersymmetry at the ATLAS experiment", *Physics Reports*, vol. 1116, pp. 261–300, 2025.
10. ATLAS Collaboration, "Characterising the Higgs boson with ATLAS data from the LHC Run-2", *Physics Reports*, vol. 1116, pp. 4–56, 2025.
11. ATLAS Collaboration, "Climbing to the Top of the ATLAS 13 TeV data", *Physics Reports*, vol. 1116, pp. 127–183, 2025.
12. ATLAS Collaboration, "Software and computing for Run 3 of the ATLAS experiment at the LHC", *The European Physical Journal C*, vol. 85, No. 3, pp. 234, 2025.
13. ATLAS Collaboration, "A search for R-parity-violating supersymmetry in final states containing many jets in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector", *Journal of High Energy Physics*, vol. 2024, No. 5, pp. 003, 2024.
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15. ATLAS Collaboration, "Combination of measurements of the top quark mass from data collected by the ATLAS and CMS experiments at $\sqrt{s} = 7$ and 8 TeV", *Physical Review Letters*, vol. 132, No. 26, pp. 261902, 2024.
16. ATLAS Collaboration, "Evidence for the $VH, H \to \tau\tau$ process with the ATLAS detector in Run 2", *Physics Letters B*, vol. 855, pp. 138817, 2024.
17. ATLAS Collaboration, "Fiducial and differential cross-section measurements of electroweak $W\gamma jj$ production in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector", *The European Physical Journal C*, vol. 84, No. 10, pp. 1064, 2024.
18. ATLAS Collaboration, "Improving topological cluster reconstruction using calorimeter cell timing in ATLAS", *The European Physical Journal C*, vol. 84, No. 5, pp. 455, 2024.
19. ATLAS Collaboration, "Measurements of inclusive and differential cross-sections of $t\bar{t}\gamma$ production in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector", *Journal of High Energy Physics*, vol. 2024, No. 10, pp. 191, 2024.
20. ATLAS Collaboration, "Operation and performance of the ATLAS tile calorimeter in LHC Run 2", *The European Physical Journal C*, vol. 84, No. 12, pp. 1313, 2024.
21. ATLAS Collaboration, "Precise measurements of W- and Z-boson transverse momentum spectra with the ATLAS detector using pp collisions at $\sqrt{s} = 5.02$ TeV and 13 TeV", *The European Physical Journal C*, vol. 84, No. 10, pp. 1126, 2024.

22. ATLAS Collaboration, "Search for nearly mass-degenerate higgsinos using low-momentum mildly displaced tracks in *pp* collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector", *Physical Review Letters*, vol. 132, No. 22, pp. 221801, 2024.
23. ATLAS Collaboration, "Search for a new Z' gauge boson via the $pp \to W^{\pm(*)} \to Z'\mu^{\pm}\nu \to \mu^{\pm}\mu^{\mp}\mu^{\pm}\nu$ process in *pp* collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector", *Physical Review D*, vol. 110, No. 7, pp. 072008, 2024.
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27. ATLAS Collaboration, "Search for new particles in final states with a boosted top quark and missing transverse momentum in proton-proton collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector", *Journal of High Energy Physics*, vol. 2024, No. 5, pp. 263, 2024.
28. ATLAS Collaboration, "Search for non-resonant production of semi-visible jets using Run 2 data in ATLAS", *Physics Letters B*, vol. 848, pp. 138324, 2024.
29. ATLAS Collaboration, "Search for pair production of higgsinos in events with two Higgs bosons and missing transverse momentum in $\sqrt{s} = 13$ TeV *pp* collisions at the ATLAS experiment", *Physical Review D*, vol. 109, No. 11, pp. 112011, 2024.
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32. ATLAS Collaboration, "Search for the non-resonant production of Higgs boson pairs via gluon fusion and vector-boson fusion in the $b\bar{b}\tau^{+}\tau^{-}$ final state in proton–proton collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector", *Physical Review D*, vol. 110, No. 3, pp. 032012, 2024.
33. ATLAS Collaboration, "Search for top-philic heavy resonances in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector", *The European Physical Journal C*, vol. 84, No. 2, pp. 157, 2024.
34. ATLAS Collaboration, "Statistical combination of ATLAS Run 2 searches for charginos and neutralinos at the LHC", *Physical Review Letters*, vol. 133, No. 3, pp. 031802, 2024.
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53. ATLAS Collaboration, "Measurement and interpretation of same-sign W boson pair production in association with two jets in *pp* collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector", *Journal of High Energy Physics*, vol. 2024, No. 4, 2024.
54. ATLAS Collaboration, "Measurement of the Z boson invisible width at $\sqrt{s} = 13$ TeV with the ATLAS detector", *Physics Letters B*, vol. 854, pp. 138705, 2024.
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81. ATLAS Collaboration, "Measurement of the $H \to \gamma\gamma$ and $H \to ZZ^* \to 4\ell$ cross-sections in pp collisions at $\sqrt{s} = 13.6$ TeV with the ATLAS detector", *The European Physical Journal C*, vol. 84, No. 1, pp. 78, 2024.
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PATENTS

1. **"Dynamically Forecasting High Resolution Air Temperature in Real-Time Using Multiple Sources"**

   Inventors: Akhalwaya, Ismail Yunus and Vos, Etienne Eben and Makhanya, Sibusisiwe Audrey and Govindasamy, Tamara Rosemary and Patel, Zubeida and Morake, Dedricks Monyai and Mahlasi, Craig and Bhamjee, Muaaz.
   Inventors on behalf of IBM Corp. South Africa: Granted 30 July 2024, ZA2023/09448 (ID: 99151468). USA: Pending, US18/200,050. Link

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